

5.1 List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Component	Description (1–2 lines)
Motherboard	The main circuit board that connects and allows communication between all computer components.
CPU (Processor)	The brain of the computer that performs calculations and executes instructions.
CPU Cooler (Fan & Heatsink)	Keeps the processor cool by removing excess heat during operation.
RAM (Memory)	Temporarily stores data and programs that the CPU is currently using for faster access.
Storage Drive (SSD/HDD)	Stores the operating system, applications, and user files permanently.
Graphics Card (GPU)	Processes graphics and video output for gaming, editing, and other visual tasks.
Power Supply Unit (PSU)	Converts AC power into DC power and supplies electricity to all PC components.
Case Fans	Improve airflow inside the cabinet by bringing in cool air and exhausting hot air.
Expansion Cards	Add extra features such as Wi-Fi, Bluetooth, sound, or additional USB ports.
SATA/Data & Power Cables	Connect storage drives and other components to the motherboard and power supply.
CMOS Battery	Maintains the BIOS settings and system clock even when the computer is turned off.

5.2 Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.



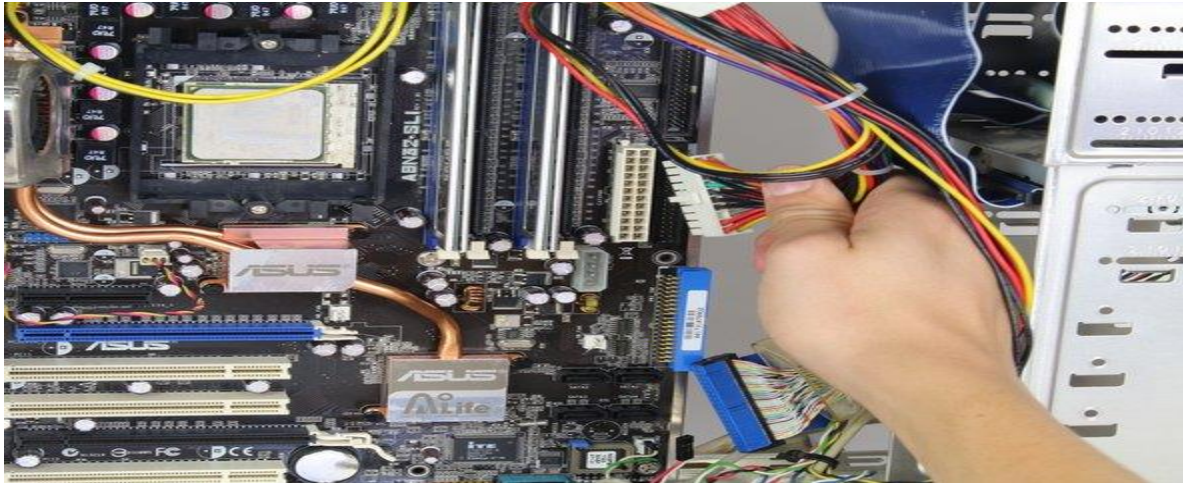
unplugging power :-

First, I shut down the desktop computer and unplugged the power cable from the power supply unit (PSU). This ensured the system was completely disconnected from electricity before opening the cabinet, making the process safe.



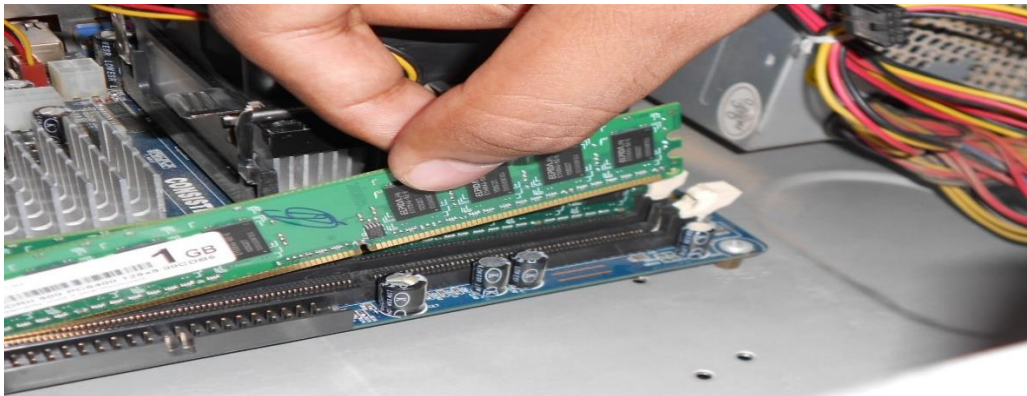
removing side panel :-

Next, I removed the screws from the back of the PC cabinet and carefully slid off the side panel. This allowed me to access all the internal components of the desktop safely.



disconnecting cables:-

After opening the cabinet, I carefully disconnected the power and data cables connected to the motherboard, storage drives, and other components. I handled the cables gently to avoid damaging the connectors.



removing RAM :-

I pressed the retaining clips on both ends of the RAM slot and carefully lifted the RAM module out. I handled it by the edges to avoid touching the gold contacts and prevent static damage.



Removing the HDD/SSD:-

I disconnected the SATA data cable and power cable from the HDD/SSD, then unscrewed or released the drive from its mounting bracket. After that, I carefully removed the storage drive from the cabinet without applying excessive force.



Removing the Motherboard:-

After disconnecting all cables and removing the attached components, I unscrewed the motherboard from the cabinet standoffs. Then, I carefully lifted it out while holding it by the edges to avoid damaging the electronic circuits.

5.3 Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Step 1: Install the Motherboard

I carefully placed the motherboard back into the cabinet, aligned it with the standoffs, and secured it with screws. This is done first because all other components connect to the motherboard.

Step 2: Install the HDD/SSD

I mounted the HDD/SSD in its drive bay and connected the SATA data cable and power cable. The storage drive must be connected so the computer can access the operating system and files.

Step 3: Install the RAM

I aligned the RAM module with the memory slot and pressed it firmly until the retaining clips locked into place. RAM must be installed before starting the computer so it can run properly.

Step 4: Reconnect All Cables

I reconnected the power supply cables, SATA cables, and front panel connectors to the motherboard and other components. Proper cable connections ensure every component receives power and communicates correctly.

Step 5: Close the Side Panel

I placed the side panel back onto the cabinet and tightened the screws. This protects the internal components and helps maintain proper airflow.

Step 6: Connect Power and Test the PC

Finally, I plugged in the power cable and turned on the computer to verify that all components were detected and the system booted successfully. This confirms the reassembly was completed correctly.

5.4 Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Safety Precaution / Tool	Why It Is Important
Turn off the PC	Prevents damage to components and avoids electrical hazards.
Unplug the power cable	Ensures there is no electricity flowing through the system.
Press the power button for a few seconds	Discharges any remaining electrical power in the PC.
Wear an anti-static wrist strap (if available)	Protects sensitive components from static electricity.
Work on a clean, flat surface	Prevents parts from falling, getting dirty, or being damaged.
Handle components by the edges	Avoids touching electronic circuits and connectors directly.
Keep screws in a small container	Prevents screws from getting lost during disassembly.
Use the correct screwdriver	Helps remove and install screws safely without damaging them.
Avoid using excessive force	Prevents damage to connectors, slots, and other components.
Organize cables properly	Makes reassembly easier and improves airflow inside the PC.